

Pharma QMS Role-Based Workflow and UAT Testing Guide

Complete role matrix, process flows, workflow handoffs, UI test cases, examples, API/BPMN mapping, and validation-ready UAT guidance

Document Version: 1.0

Prepared Date: 2026-06-23

System Scope: Complete Pharma eQMS with CAPA, Deviations, Change Control, Document Management, Training, Risk, Audit, Supplier Quality, Complaints, Nonconformance, Equipment, Calibration, Validation, Management Review, Regulatory Commitments, Reporting, e-signatures, audit trails, RBAC, Flowable workflow, and GCS document storage.

1. Document Purpose

This document defines the main Pharma QMS user flows that must be tested before customer release or production go-live. It is written for QA Managers, Validation Engineers, Business Analysts, CSV teams, auditors, implementation consultants, and product teams. The focus is role-based workflow testing: one user creates a quality record, the next user reviews it, approvers electronically approve it, and related downstream records such as CAPA, Change Control, Documents, Training, Risk Assessments, Audits, Complaints, Supplier CAPAs, and Management Review actions are created and closed with full traceability.

- Use this document as a UAT baseline.
- Adapt role names and permission codes to your exact implementation.
- Map each UI screen and backend API endpoint to the relevant test case.
- Attach screenshots and evidence during execution for validation records.

2. Primary Business Scenario Used Throughout the Document

Scenario: During production of Batch B-2026-001 for Product PX-100, an operator observes that the granulation room temperature exceeded the approved range for 42 minutes. The operator raises a Deviation. QA performs investigation and risk assessment. The root cause is linked to overdue HVAC calibration and unclear SOP instructions. QA raises a CAPA, creates a Change Control to revise the SOP and calibration frequency, routes the revised SOP through Document Management, assigns Training to affected users, and verifies CAPA effectiveness after 30 days. The issue is later reviewed in Management Review and checked during an internal audit.

Record	Example ID	Purpose
Deviation	DEV-2026-0001	Capture temperature excursion and initial investigation
Risk Assessment	RISK-2026-0001	Evaluate impact to product quality and patient safety
CAPA	CAPA-2026-0001	Correct root cause and prevent recurrence
Change Control	CC-2026-0001	Revise SOP and calibration frequency
Document	SOP-MFG-001 v3.0	Controlled procedure update
Training	TRN-2026-0001	Assign retraining to operators and supervisors
Audit Finding	AUD-F-2026-0001	Verify process compliance during internal audit
Management Review Action	MR-ACT-2026-0001	Track trend and management decision

3. User Types, Roles, and Permissions

Role	Main Responsibilities	Typical Permissions
System Administrator	Configure users, roles, permissions, master data, workflows, system settings, and integrations.	Full admin; no business approval unless also assigned QA role.
Initiator / Operator	Create events such as deviations, complaints, nonconformances, and observations.	Create draft, upload evidence, submit, view own records.
QA Investigator	Perform investigation, root cause analysis, impact assessment, and proposed actions.	Edit assigned investigations, create linked CAPA/risk, submit for review.
QA Reviewer	Review completeness, classify severity, request changes, approve investigation package.	Review, return, approve review step, add comments.
QA Approver / QA Manager	Final approval for deviations, CAPAs, changes, documents, audits, and risk decisions.	Approve/reject, e-sign, close records, assign effectiveness checks.
CAPA Owner	Execute corrective/preventive actions and provide evidence.	Update assigned CAPA tasks, attach evidence, mark complete.
Change Owner	Coordinate impact assessment, implementation tasks, verification, and closure.	Create/update change records, manage tasks, submit closure.
Document Author	Create and revise SOPs, policies, forms, and controlled documents.	Draft document, upload file, route review.
Document Controller	Control document numbering, versioning, release, obsolescence, and GCS file lifecycle.	Approve metadata, issue/effective, obsolete, publish.
Trainer / Training Coordinator	Assign training, monitor completion, manage curriculum and retraining.	Create assignments, track completion, escalate overdue.
Trainee	Complete assigned training and acknowledgements.	Read controlled documents, complete quiz/acknowledge.
Risk Owner	Own risk record, define controls, assess residual risk.	Create/edit risk, propose mitigation, submit approval.
Auditor	Plan and execute audits, record observations/findings.	Create audit plans, checklists, findings, reports.
Supplier Quality Manager	Qualify suppliers, manage supplier audits, supplier CAPAs and scorecards.	Create/update supplier records, approve supplier status.
Complaint Handler	Intake complaints, triage, collect customer/product/batch data.	Create complaint, link product/batch, submit investigation.
Validation Engineer	Create validation plans, requirements, protocols, execute tests, collect evidence.	Create validation project, execute tests, attach evidence, route approvals.
Maintenance / Calibration User	Maintain equipment, calibration and PM records.	Create/update equipment events, upload certificates, close calibration tasks.
Executive / Management Reviewer	Review quality metrics and approve management review actions.	View dashboards, approve management review minutes/actions.
Regulatory Affairs User	Track regulatory observations, commitments, responses, and agency actions.	Create/update regulatory commitments, link CAPA/changes, submit responses.

4. Permission Matrix for UI and API Testing

Use this matrix to verify RBAC behavior. Each permission should be tested from UI and backend APIs. Unauthorized access should return 401 or 403 and should also be captured in security logs if implemented.

Permission Area	Initiator	Investigator	Reviewer	Approver	Admin	Auditor	Doc Control	Trainer	Executive
Create Record	Yes	Yes	Limited	Yes	Yes	Yes	Yes	Yes	No
Edit Draft	Own	Assigned	No	No	Yes	Assigned	Assigned	Assigned	No
Submit Workflow	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	No
Review Task	No	Assigned	Yes	Yes	Yes	Assigned	Yes	Yes	No
Approve / E-sign	No	No	Limited	Yes	Configurable	Limited	Yes	Limited	Management Review only
Close Record	No	No	No	Yes	Yes	Limited	Limited	No	No
Delete / Void	No	No	No	Limited	Configurable	No	No	No	No
View Audit Trail	Own	Assigned	Yes	Yes	Yes	Yes	Yes	Yes	Read-only
Reports/Dashboards	Limited	Department	Department	All QA	All	Audit	Documents	Training	Management KPIs

5. Main Pharma QMS Flows to Test

The following flows are the most important end-to-end flows for a complete Pharma QMS. Each flow validates workflow routing, user role separation, audit trail generation, e-signature enforcement, notifications, linked records, reporting, and closure rules.

1. Deviation to CAPA to Change Control to Document to Training

Operator creates Deviation -> QA Investigator investigates -> QA Reviewer reviews -> QA Manager approves -> CAPA is created -> CAPA Owner executes actions -> Change Control is initiated -> SOP is revised -> Document Controller issues new SOP -> Training is assigned -> Trainees complete -> Effectiveness verified -> Closure.

2. Complaint to Investigation to CAPA

Complaint Handler creates complaint -> Product/batch linked -> QA triage -> Investigation -> risk/medical escalation if required -> CAPA created -> response generated -> closure.

3. Audit Finding to CAPA

Auditor schedules audit -> executes checklist -> records finding -> classifies critical/major/minor -> QA Manager approves finding -> CAPA generated -> effectiveness verified -> audit closed.

4. Supplier Qualification to Supplier CAPA

Supplier Quality creates supplier -> qualification checklist -> risk rating -> approval -> supplier audit -> finding -> supplier CAPA -> requalification.

5. Risk Assessment to Mitigation

Risk Owner creates risk -> scores severity/occurrence/detectability -> proposes mitigation -> residual risk -> QA approval -> linked actions.

6. Equipment Calibration to Deviation

Calibration due -> task assigned -> missed/failed calibration -> deviation created -> impact assessment -> CAPA/change if needed.

7. Validation Project Lifecycle

Validation Engineer creates URS -> FS/DS -> IQ/OQ/PQ protocols -> execute tests -> deviations for failures -> traceability matrix -> validation report approval.

8. Management Review

Metrics aggregated -> management review meeting -> CAPA trends/audit trends/complaint trends reviewed -> actions assigned -> minutes approved.

6.1 Deviation Management Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Deviation Management Flow. Scope: Deviation record, attachments, classification, investigation, root cause, risk, CAPA decision, closure.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Initiator	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Initiator	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	QA Investigator	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	QA Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	QA Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.2 CAPA Management Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for CAPA Management Flow. Scope: CAPA plan, action tasks, due dates, evidence, effectiveness check, closure.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	QA Investigator	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	QA Investigator	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	CAPA Owner	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	QA Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	QA Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.3 Change Control Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Change Control Flow. Scope: Change request, impact assessment, risk, implementation, verification, closure.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Change Owner	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Change Owner	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	Impact Assessors	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Change Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Change Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.4 Document Management Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Document Management Flow. Scope: Draft, review, approval, version control, effective date, obsolescence.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Document Author	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Document Author	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	Document Reviewer	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	Document Controller	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	QA Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	QA Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.5 Training Management Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Training Management Flow. Scope: Training assignment, completion, quiz/acknowledgement, overdue escalation.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Training Coordinator	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Training Coordinator	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	Trainee	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	Supervisor	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Training Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Training Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.6 Risk Management Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Risk Management Flow. Scope: Risk register, assessment, mitigation, residual risk, acceptance.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Risk Owner	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Risk Owner	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	SME Reviewer	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Risk Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Risk Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.7 Audit Management Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Audit Management Flow. Scope: Audit plan, checklist, findings, report, CAPA linkage.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Auditor	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Auditor	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	Auditee	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Audit Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Audit Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.8 Complaint Management Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Complaint Management Flow. Scope: Complaint intake, product/batch link, investigation, escalation, CAPA, closure.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Complaint Handler	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Complaint Handler	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	QA Investigator	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Complaint Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Complaint Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.9 Supplier Quality Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Supplier Quality Flow. Scope: Supplier qualification, ASL, scorecard, audit, supplier CAPA.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Supplier Quality Owner	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Supplier Quality Owner	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	Supplier Reviewer	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Approver	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Supplier Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Supplier Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.10 Nonconformance Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Nonconformance Flow. Scope: Material/product NC, hold, disposition, rework/reject/release.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Warehouse/Manufacturing User	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Warehouse/Manufacturing User	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	QA Investigator	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Disposition Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Disposition Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.11 Equipment and Calibration Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Equipment and Calibration Flow. Scope: Equipment master, calibration, PM, qualification, asset history.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Maintenance User	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Maintenance User	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	Calibration Reviewer	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Equipment Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Equipment Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.12 Validation Management Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Validation Management Flow. Scope: URS, FS, DS, IQ, OQ, PQ, traceability, validation report.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Validation Engineer	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Validation Engineer	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	Tester	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	QA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Validation Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Validation Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.13 Management Review Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Management Review Flow. Scope: Metrics pack, meeting agenda, minutes, actions, approval.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	QA Metrics Owner	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	QA Metrics Owner	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	Functional Leaders	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	Executive Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	Management Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	Management Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

6.14 Regulatory Commitments Flow

This flow tests creation, review, approval, electronic signature, audit trail, notifications, record linking, and closure for Regulatory Commitments Flow. Scope: Inspection observations, commitments, agency responses, action plans.

Workflow Step	Responsible Role	Expected UI Behavior	Expected Backend/Flowable Behavior
Create/Draft	Regulatory User	User opens module, clicks Create, completes required fields, saves draft.	Record created with status Draft. Audit trail created. Flowable instance may start on Submit.
Submit	Regulatory User	Submit button validates mandatory fields and shows confirmation.	Status changes to Submitted. Flowable task assigned to next role. Notification generated.
Review/Work	QA Owner	Assigned user opens My Tasks, completes investigation/action fields, attaches evidence.	Task completed. Variables updated. Attachments stored in GCS. Audit trail updated.
QA Review	RA Reviewer	Reviewer can approve, reject, or request more information.	Gateway routes to rework or approval. Comments and decisions persisted.
Final Approval	RA/QA Approver	Approver reviews final package and performs e-signature.	E-signature record created. Status moves to Approved/Closed or next stage.
Closure/Linking	RA/QA Approver	System shows linked records and closure checklist.	Closure rules validated. Metrics updated. Search index and reporting updated.

Example: Use the primary business scenario unless the module needs a specific example. Verify that users without the assigned role cannot perform the next workflow action. Verify that the audit trail captures every status change, field change, attachment upload, reassignment, approval, rejection, and signature.

7. Flowable BPMN Workflow Mapping

Each module should have a BPMN process definition in Flowable. Test both happy path and exception path. Typical elements are Start Event, User Tasks, Service Tasks, Exclusive Gateways, Timer Events, Boundary Events, Notification Service Tasks, and End Event. The frontend should use task APIs to display available actions based on current user, role, and record status.

BPMN Element	QMS Usage	Test Expectation
Start Event	Submit record starts workflow.	Process instance created with business key equal to QMS record number.
User Task	Review, approval, investigation, action execution.	Task visible only to assigned user/group.
Service Task	Send notification, generate report, update metrics.	Service executes automatically and logs success/failure.
Exclusive Gateway	CAPA required? Risk acceptable? Change approved?	Correct branch selected based on process variables.
Timer Event	Due date reminders and escalations.	Notification sent before due date; escalation after due date.
Boundary Event	Rejection/rework paths.	Record returns to previous owner with comments.
End Event	Workflow closed.	Status becomes Closed/Effective/Approved and record is locked.

8. Backend API Mapping for Testing

For every UI test, validate the corresponding backend API behavior. Test positive cases, negative cases, validation errors, unauthorized access, workflow state restrictions, audit trail creation, and idempotency where applicable.

Module	Create	Read	Update	Workflow Action	Linked Records
deviations	POST /api/deviations	GET /api/deviations/{id}	PUT /api/deviations/{id}	POST /api/deviations/{id}/actions/{action}	GET /api/deviations/{id}/links
capa	POST /api/capa	GET /api/capa/{id}	PUT /api/capa/{id}	POST /api/capa/{id}/actions/{action}	GET /api/capa/{id}/links
change-controls	POST /api/change-controls	GET /api/change-controls/{id}	PUT /api/change-controls/{id}	POST /api/change-controls/{id}/actions/{action}	GET /api/change-controls/{id}/links
documents	POST /api/documents	GET /api/documents/{id}	PUT /api/documents/{id}	POST /api/documents/{id}/actions/{action}	GET /api/documents/{id}/links
training	POST /api/training	GET /api/training/{id}	PUT /api/training/{id}	POST /api/training/{id}/actions/{action}	GET /api/training/{id}/links
risks	POST /api/risks	GET /api/risks/{id}	PUT /api/risks/{id}	POST /api/risks/{id}/actions/{action}	GET /api/risks/{id}/links
audits	POST /api/audits	GET /api/audits/{id}	PUT /api/audits/{id}	POST /api/audits/{id}/actions/{action}	GET /api/audits/{id}/links
complaints	POST /api/complaints	GET /api/complaints/{id}	PUT /api/complaints/{id}	POST /api/complaints/{id}/actions/{action}	GET /api/complaints/{id}/links
suppliers	POST /api/suppliers	GET /api/suppliers/{id}	PUT /api/suppliers/{id}	POST /api/suppliers/{id}/actions/{action}	GET /api/suppliers/{id}/links
nonconformances	POST /api/nonconformances	GET /api/nonconformances/{id}	PUT /api/nonconformances/{id}	POST /api/nonconformances/{id}/actions/{action}	GET /api/nonconformances/{id}/links
equipment	POST /api/equipment	GET /api/equipment/{id}	PUT /api/equipment/{id}	POST /api/equipment/{id}/actions/{action}	GET /api/equipment/{id}/links
calibrations	POST /api/calibrations	GET /api/calibrations/{id}	PUT /api/calibrations/{id}	POST /api/calibrations/{id}/actions/{action}	GET /api/calibrations/{id}/links
validation-projects	POST /api/validation-projects	GET /api/validation-projects/{id}	PUT /api/validation-projects/{id}	POST /api/validation-projects/{id}/actions/{action}	GET /api/validation-projects/{id}/links
management-reviews	POST /api/management-reviews	GET /api/management-reviews/{id}	PUT /api/management-reviews/{id}	POST /api/management-reviews/{id}/actions/{action}	GET /api/management-reviews/{id}/links
regulatory-commitments	POST /api/regulatory-commitments	GET /api/regulatory-commitments/{id}	PUT /api/regulatory-commitments/{id}	POST /api/regulatory-commitments/{id}/actions/{action}	GET /api/regulatory-commitments/{id}/links
esignatures	POST /api/esignatures	GET /api/esignatures/{id}	PUT /api/esignatures/{id}	POST /api/esignatures/{id}/actions/{action}	GET /api/esignatures/{id}/links
audit-trails	POST /api/audit-trails	GET /api/audit-trails/{id}	PUT /api/audit-trails/{id}	POST /api/audit-trails/{id}/actions/{action}	GET /api/audit-trails/{id}/links
notifications	POST /api/notifications	GET /api/notifications/{id}	PUT /api/notifications/{id}	POST /api/notifications/{id}/actions/{action}	GET /api/notifications/{id}/links
reports	POST /api/reports	GET /api/reports/{id}	PUT /api/reports/{id}	POST /api/reports/{id}/actions/{action}	GET /api/reports/{id}/links
master-data	POST /api/master-data	GET /api/master-data/{id}	PUT /api/master-data/{id}	POST /api/master-data/{id}/actions/{action}	GET /api/master-data/{id}/links

9. UAT Test Case Structure

Every test case should be executed with a named test user, expected role, preconditions, data set, steps, expected result, actual result, evidence, and pass/fail status. Attach screenshots, downloaded reports, audit trail exports, e-signature evidence, and API logs where required.

Field	Description
Test ID	Unique identifier such as UAT-DEV-001
Title	Short test objective
Role/User	User role used for login
Preconditions	Master data, roles, workflow config, existing records
Steps	Detailed UI actions
Expected Result	System result and validation evidence
Evidence	Screenshot, report, audit trail, API response
Pass/Fail	Execution result

10.1 UAT Test Cases - Deviation Management

The following test cases validate Deviation Management from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-DEV-001: Create Draft

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Initiator / QA Investigator. Open Deviation Management. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-002: Submit Workflow

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-003: Reviewer Approval

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-004: Reject/Rework Path

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-005: Final E-sign Approval

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-006: Attachment Upload

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-007: Linked Record Creation

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-008: Overdue Escalation

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-009: Audit Trail Review

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DEV-010: Report/Dashboard Update

Field	Content
Role/User	Initiator / QA Investigator
Preconditions	Deviation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.2 UAT Test Cases - CAPA Management

The following test cases validate CAPA Management from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-CAP-001: Create Draft

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as QA Investigator / CAPA Owner. Open CAPA Management. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-002: Submit Workflow

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-003: Reviewer Approval

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-004: Reject/Rework Path

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-005: Final E-sign Approval

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-006: Attachment Upload

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-007: Linked Record Creation

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-008: Overdue Escalation

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-009: Audit Trail Review

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CAP-010: Report/Dashboard Update

Field	Content
Role/User	QA Investigator / CAPA Owner
Preconditions	CAPA Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.3 UAT Test Cases - Change Control

The following test cases validate Change Control from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-CHA-001: Create Draft

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Change Owner. Open Change Control. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-002: Submit Workflow

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-003: Reviewer Approval

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-004: Reject/Rework Path

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-005: Final E-sign Approval

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-006: Attachment Upload

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-007: Linked Record Creation

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-008: Overdue Escalation

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-009: Audit Trail Review

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-CHA-010: Report/Dashboard Update

Field	Content
Role/User	Change Owner
Preconditions	Change Control module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.4 UAT Test Cases - Document Management

The following test cases validate Document Management from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-DOC-001: Create Draft

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Document Author / Document Controller. Open Document Management. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-002: Submit Workflow

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-003: Reviewer Approval

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-004: Reject/Rework Path

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-005: Final E-sign Approval

Field	Content
Role/User	Document Author / Document Controller

Field	Content
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-006: Attachment Upload

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-007: Linked Record Creation

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-008: Overdue Escalation

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-009: Audit Trail Review

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-DOC-010: Report/Dashboard Update

Field	Content
Role/User	Document Author / Document Controller
Preconditions	Document Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.5 UAT Test Cases - Training Management

The following test cases validate Training Management from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-TRA-001: Create Draft

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Training Coordinator / Trainee. Open Training Management. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-002: Submit Workflow

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-003: Reviewer Approval

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-004: Reject/Rework Path

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-005: Final E-sign Approval

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-006: Attachment Upload

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-007: Linked Record Creation

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-008: Overdue Escalation

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-009: Audit Trail Review

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-TRA-010: Report/Dashboard Update

Field	Content
Role/User	Training Coordinator / Trainee
Preconditions	Training Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.6 UAT Test Cases - Risk Management

The following test cases validate Risk Management from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-RIS-001: Create Draft

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Risk Owner. Open Risk Management. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-002: Submit Workflow

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-003: Reviewer Approval

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-004: Reject/Rework Path

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-005: Final E-sign Approval

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-006: Attachment Upload

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-007: Linked Record Creation

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-008: Overdue Escalation

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-009: Audit Trail Review

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-RIS-010: Report/Dashboard Update

Field	Content
Role/User	Risk Owner
Preconditions	Risk Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.7 UAT Test Cases - Audit Management

The following test cases validate Audit Management from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-AUD-001: Create Draft

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Auditor. Open Audit Management. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-002: Submit Workflow

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-003: Reviewer Approval

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-004: Reject/Rework Path

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-005: Final E-sign Approval

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-006: Attachment Upload

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-007: Linked Record Creation

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-008: Overdue Escalation

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-009: Audit Trail Review

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-AUD-010: Report/Dashboard Update

Field	Content
Role/User	Auditor
Preconditions	Audit Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.8 UAT Test Cases - Complaint Management

The following test cases validate Complaint Management from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-COM-001: Create Draft

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Complaint Handler. Open Complaint Management. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-002: Submit Workflow

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-003: Reviewer Approval

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-004: Reject/Rework Path

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-005: Final E-sign Approval

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-006: Attachment Upload

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-007: Linked Record Creation

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-008: Overdue Escalation

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-009: Audit Trail Review

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-COM-010: Report/Dashboard Update

Field	Content
Role/User	Complaint Handler
Preconditions	Complaint Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.9 UAT Test Cases - Supplier Quality

The following test cases validate Supplier Quality from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-SUP-001: Create Draft

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Supplier Quality Manager. Open Supplier Quality. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-002: Submit Workflow

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-003: Reviewer Approval

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-004: Reject/Rework Path

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-005: Final E-sign Approval

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-006: Attachment Upload

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-007: Linked Record Creation

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-008: Overdue Escalation

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-009: Audit Trail Review

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-SUP-010: Report/Dashboard Update

Field	Content
Role/User	Supplier Quality Manager
Preconditions	Supplier Quality module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.10 UAT Test Cases - Nonconformance

The following test cases validate Nonconformance from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-NON-001: Create Draft

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Warehouse or Manufacturing User. Open Nonconformance. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-002: Submit Workflow

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-003: Reviewer Approval

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-004: Reject/Rework Path

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-005: Final E-sign Approval

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-006: Attachment Upload

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-007: Linked Record Creation

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-008: Overdue Escalation

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-009: Audit Trail Review

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-NON-010: Report/Dashboard Update

Field	Content
Role/User	Warehouse or Manufacturing User
Preconditions	Nonconformance module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.11 UAT Test Cases - Equipment and Calibration

The following test cases validate Equipment and Calibration from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-EQU-001: Create Draft

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Maintenance User. Open Equipment and Calibration. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-002: Submit Workflow

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-003: Reviewer Approval

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-004: Reject/Rework Path

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-005: Final E-sign Approval

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-006: Attachment Upload

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-007: Linked Record Creation

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-008: Overdue Escalation

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-009: Audit Trail Review

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-EQU-010: Report/Dashboard Update

Field	Content
Role/User	Maintenance User
Preconditions	Equipment and Calibration module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.12 UAT Test Cases - Validation Management

The following test cases validate Validation Management from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-VAL-001: Create Draft

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Validation Engineer. Open Validation Management. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-002: Submit Workflow

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-003: Reviewer Approval

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-004: Reject/Rework Path

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-005: Final E-sign Approval

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-006: Attachment Upload

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-007: Linked Record Creation

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-008: Overdue Escalation

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-009: Audit Trail Review

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-VAL-010: Report/Dashboard Update

Field	Content
Role/User	Validation Engineer
Preconditions	Validation Management module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.13 UAT Test Cases - Management Review

The following test cases validate Management Review from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-MAN-001: Create Draft

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Executive / QA Manager. Open Management Review. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-002: Submit Workflow

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-003: Reviewer Approval

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-004: Reject/Rework Path

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-005: Final E-sign Approval

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-006: Attachment Upload

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-007: Linked Record Creation

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-008: Overdue Escalation

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-009: Audit Trail Review

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-MAN-010: Report/Dashboard Update

Field	Content
Role/User	Executive / QA Manager
Preconditions	Management Review module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

10.14 UAT Test Cases - Regulatory Commitments

The following test cases validate Regulatory Commitments from UI, workflow, backend, audit trail, e-signature, and reporting perspectives.

UAT-REG-001: Create Draft

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as Regulatory Affairs User. Open Regulatory Commitments. Click Create. Enter required fields. Save draft.
Expected Result	Draft is saved, record number is generated, mandatory validation works, audit trail records creation.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-002: Submit Workflow

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Open draft record. Complete mandatory fields. Click Submit.
Expected Result	Status changes to Submitted, Flowable task assigned to reviewer, notification generated.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-003: Reviewer Approval

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Open My Tasks. Review record. Add comments. Approve.
Expected Result	Task completed, audit trail records review decision, next task routed correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-004: Reject/Rework Path

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Login as reviewer. Reject record with reason.
Expected Result	Record returns to creator/owner with comments, status shows Rework, audit trail captures reason.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-005: Final E-sign Approval

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.

Field	Content
UI Steps	Login as approver. Click Approve. Enter credentials/re-authentication and signature meaning.
Expected Result	Signature manifestation is stored, record status updates, approver name/date/reason visible.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-006: Attachment Upload

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Upload evidence file from UI. Download/open it from record.
Expected Result	File stored in GCS, metadata recorded, permissions enforced, audit trail captures upload.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-007: Linked Record Creation

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Create linked CAPA/Risk/Change/Training as applicable.
Expected Result	Linked record created with source reference and visible in both records.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-008: Overdue Escalation

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Set due date to trigger reminder/escalation.
Expected Result	Notification sent and escalation task created based on SLA rules.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-009: Audit Trail Review

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Modify a field and view audit trail.
Expected Result	Audit trail shows who, what, when, old value, new value, and reason if required.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

UAT-REG-010: Report/Dashboard Update

Field	Content
Role/User	Regulatory Affairs User
Preconditions	Regulatory Commitments module enabled; user has required role; master data exists; Flowable process deployed.
UI Steps	Close record and refresh dashboard/report.
Expected Result	Metric counters and trend charts update correctly.
Evidence Required	Screenshot of record, workflow history, audit trail, e-signature if applicable, notification, report/API response.

11. End-to-End UAT Script: Deviation to CAPA to Change Control to Document to Training

This is the highest priority test flow because it proves that the QMS behaves as an integrated system rather than separate modules.

Step	Login Role	Action	Expected Result
1	Operator	Create Deviation DEV-2026-0001 for temperature excursion.	Deviation saved and submitted.
2	QA Investigator	Perform investigation and root cause analysis.	Investigation completed; risk required.
3	Risk Owner	Create and approve Risk RISK-2026-0001.	Initial and residual risk calculated.
4	QA Manager	Approve deviation investigation and require CAPA.	CAPA creation enabled.
5	CAPA Owner	Create CAPA with corrective and preventive actions.	CAPA tasks assigned.
6	Change Owner	Create Change Control to revise SOP and calibration frequency.	Change impact assessment started.
7	Document Author	Revise SOP-MFG-001 to version 3.0.	Document routed for review.
8	Document Controller	Approve and issue SOP.	New version becomes Effective.
9	Training Coordinator	Assign training to operators and supervisors.	Training assignments generated.
10	Trainees	Complete training acknowledgements/quiz.	Training compliance updated.
11	QA Manager	Verify CAPA effectiveness after monitoring period.	CAPA closed with e-signature.
12	Executive	Review trend in Management Review.	Management action documented.

12. Master Data Setup Guide

Before UAT execution, configure controlled master data. Each master data record should have owner, status, effective date, audit trail, and change control requirements where applicable.

Master Data	Example Values	Used By
Sites	Site A, Site B	All modules, RBAC, reports
Products	PX-100, PX-200	Deviation, complaint, batch, risk
Materials	RM-1001, API-001	Nonconformance, supplier quality
Batches	B-2026-001	Deviation, complaint, LIMS/MES links
Departments	Manufacturing, QA, QC, Warehouse	Tasks, routing, reports
Suppliers	Supplier A, Supplier B	Supplier quality, materials
Equipment	HVAC-001, HPLC-002	Calibration, deviation, validation
Document Types	SOP, Policy, Form, WI	Document management
Training Curricula	Manufacturing GMP, QC GMP	Training assignments
Risk Categories	Process, Product, Supplier, Equipment	Risk management

13. Part 11, Audit Trail, and E-signature Verification

This section provides scenarios that must pass for regulatory readiness. The system should enforce unique users, access controls, secure audit trails, electronic signature meaning, re-authentication, and signature manifestation.

Scenario	Test	Expected Evidence
Unique user login	Try shared/generic user or disabled user.	Access denied or policy enforced.
E-sign re-authentication	Approve a record using e-sign modal.	User identity, timestamp, meaning, record ID captured.
Signature manifestation	Open approved record and PDF/report.	Name, date/time, meaning displayed.
Audit trail old/new values	Change classification from Minor to Major.	Old value, new value, user, timestamp captured.
Reason for change	Modify approved/controlled field where reason required.	Reason mandatory and visible in audit trail.
Immutable audit trail	Attempt API update/delete audit record.	Operation denied.
Session timeout	Leave session idle past configured time.	User must re-authenticate.
RBAC enforcement	Initiator attempts final approval.	Approval action unavailable or denied.

14. IQ/OQ/PQ Validation Templates

Use IQ to verify installation and configuration, OQ to verify functional behavior under controlled test conditions, and PQ to verify real business process execution using representative users and data.

IQ - Installation Qualification

Field	Template Content
Objective	Verify IQ - Installation Qualification scope for Pharma QMS.
Preconditions	Environment, user access, master data, workflows, and integrations are prepared.
Execution Steps	Execute approved test scripts and record objective evidence.
Acceptance Criteria	All critical tests pass; deviations documented; approvals completed.
Evidence	Screenshots, audit trails, API logs, reports, approvals, validation summary.
Approval	Validation Engineer, QA, System Owner.

OQ - Operational Qualification

Field	Template Content
Objective	Verify OQ - Operational Qualification scope for Pharma QMS.
Preconditions	Environment, user access, master data, workflows, and integrations are prepared.
Execution Steps	Execute approved test scripts and record objective evidence.
Acceptance Criteria	All critical tests pass; deviations documented; approvals completed.
Evidence	Screenshots, audit trails, API logs, reports, approvals, validation summary.
Approval	Validation Engineer, QA, System Owner.

PQ - Performance Qualification

Field	Template Content
Objective	Verify PQ - Performance Qualification scope for Pharma QMS.
Preconditions	Environment, user access, master data, workflows, and integrations are prepared.
Execution Steps	Execute approved test scripts and record objective evidence.
Acceptance Criteria	All critical tests pass; deviations documented; approvals completed.
Evidence	Screenshots, audit trails, API logs, reports, approvals, validation summary.
Approval	Validation Engineer, QA, System Owner.

15. Production Deployment Checklist

The production deployment checklist confirms that the validated QMS is ready for go-live.

Area	Checklist Items
Infrastructure	Database backup, GCS bucket permissions, Flowable engines, API gateway, monitoring, logs, TLS certificates.
Security	SSO configured, RBAC verified, admin accounts reviewed, password/session policy configured.
Application	All modules enabled, workflow definitions deployed, feature flags set, search indexes created.
Validation	IQ/OQ/PQ approved, open defects assessed, validation summary report approved.
Data Migration	Master data loaded, documents migrated, user accounts imported, sample records reconciled.
Integrations	ERP/LIMS/MES/LMS/HR/IdP connectivity tested, error handling verified.
Operational Readiness	Support users trained, SOPs approved, go-live plan approved, rollback plan available.
Post Go-Live	Hypercare monitoring, daily issue review, first audit trail review, first backup restore check.

Appendix A.1: Detailed Example and Evidence Checklist - Deviation Management

This appendix page provides reusable evidence guidance for Deviation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DEV-EX-001. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.2: Detailed Example and Evidence Checklist - CAPA Management

This appendix page provides reusable evidence guidance for CAPA Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CAP-EX-002. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.3: Detailed Example and Evidence Checklist - Change Control

This appendix page provides reusable evidence guidance for Change Control. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CHA-EX-003. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.4: Detailed Example and Evidence Checklist - Document Management

This appendix page provides reusable evidence guidance for Document Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DOC-EX-004. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.5: Detailed Example and Evidence Checklist - Training Management

This appendix page provides reusable evidence guidance for Training Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: TRA-EX-005. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.6: Detailed Example and Evidence Checklist - Risk Management

This appendix page provides reusable evidence guidance for Risk Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: RIS-EX-006. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.7: Detailed Example and Evidence Checklist - Audit Management

This appendix page provides reusable evidence guidance for Audit Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: AUD-EX-007. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.8: Detailed Example and Evidence Checklist - Complaint Management

This appendix page provides reusable evidence guidance for Complaint Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: COM-EX-008. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.9: Detailed Example and Evidence Checklist - Supplier Quality

This appendix page provides reusable evidence guidance for Supplier Quality. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: SUP-EX-009. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.10: Detailed Example and Evidence Checklist - Nonconformance

This appendix page provides reusable evidence guidance for Nonconformance. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: NON-EX-010. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.11: Detailed Example and Evidence Checklist - Equipment and Calibration

This appendix page provides reusable evidence guidance for Equipment and Calibration. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: EQU-EX-011. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.12: Detailed Example and Evidence Checklist - Validation Management

This appendix page provides reusable evidence guidance for Validation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: VAL-EX-012. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.13: Detailed Example and Evidence Checklist - Management Review

This appendix page provides reusable evidence guidance for Management Review. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: MAN-EX-013. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.14: Detailed Example and Evidence Checklist - Regulatory Commitments

This appendix page provides reusable evidence guidance for Regulatory Commitments. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: REG-EX-014. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.15: Detailed Example and Evidence Checklist - Deviation Management

This appendix page provides reusable evidence guidance for Deviation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DEV-EX-015. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.16: Detailed Example and Evidence Checklist - CAPA Management

This appendix page provides reusable evidence guidance for CAPA Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CAP-EX-016. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.17: Detailed Example and Evidence Checklist - Change Control

This appendix page provides reusable evidence guidance for Change Control. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CHA-EX-017. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.18: Detailed Example and Evidence Checklist - Document Management

This appendix page provides reusable evidence guidance for Document Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DOC-EX-018. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.19: Detailed Example and Evidence Checklist - Training Management

This appendix page provides reusable evidence guidance for Training Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: TRA-EX-019. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.20: Detailed Example and Evidence Checklist - Risk Management

This appendix page provides reusable evidence guidance for Risk Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: RIS-EX-020. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.21: Detailed Example and Evidence Checklist - Audit Management

This appendix page provides reusable evidence guidance for Audit Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: AUD-EX-021. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.22: Detailed Example and Evidence Checklist - Complaint Management

This appendix page provides reusable evidence guidance for Complaint Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: COM-EX-022. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.23: Detailed Example and Evidence Checklist - Supplier Quality

This appendix page provides reusable evidence guidance for Supplier Quality. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: SUP-EX-023. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.24: Detailed Example and Evidence Checklist - Nonconformance

This appendix page provides reusable evidence guidance for Nonconformance. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: NON-EX-024. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.25: Detailed Example and Evidence Checklist - Equipment and Calibration

This appendix page provides reusable evidence guidance for Equipment and Calibration. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: EQU-EX-025. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.26: Detailed Example and Evidence Checklist - Validation Management

This appendix page provides reusable evidence guidance for Validation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: VAL-EX-026. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.27: Detailed Example and Evidence Checklist - Management Review

This appendix page provides reusable evidence guidance for Management Review. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: MAN-EX-027. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.28: Detailed Example and Evidence Checklist - Regulatory Commitments

This appendix page provides reusable evidence guidance for Regulatory Commitments. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: REG-EX-028. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.29: Detailed Example and Evidence Checklist - Deviation Management

This appendix page provides reusable evidence guidance for Deviation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DEV-EX-029. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.30: Detailed Example and Evidence Checklist - CAPA Management

This appendix page provides reusable evidence guidance for CAPA Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CAP-EX-030. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.31: Detailed Example and Evidence Checklist - Change Control

This appendix page provides reusable evidence guidance for Change Control. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CHA-EX-031. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.32: Detailed Example and Evidence Checklist - Document Management

This appendix page provides reusable evidence guidance for Document Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DOC-EX-032. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.33: Detailed Example and Evidence Checklist - Training Management

This appendix page provides reusable evidence guidance for Training Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: TRA-EX-033. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.34: Detailed Example and Evidence Checklist - Risk Management

This appendix page provides reusable evidence guidance for Risk Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: RIS-EX-034. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.35: Detailed Example and Evidence Checklist - Audit Management

This appendix page provides reusable evidence guidance for Audit Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: AUD-EX-035. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.36: Detailed Example and Evidence Checklist - Complaint Management

This appendix page provides reusable evidence guidance for Complaint Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: COM-EX-036. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.37: Detailed Example and Evidence Checklist - Supplier Quality

This appendix page provides reusable evidence guidance for Supplier Quality. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: SUP-EX-037. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.38: Detailed Example and Evidence Checklist - Nonconformance

This appendix page provides reusable evidence guidance for Nonconformance. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: NON-EX-038. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.39: Detailed Example and Evidence Checklist - Equipment and Calibration

This appendix page provides reusable evidence guidance for Equipment and Calibration. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: EQU-EX-039. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.40: Detailed Example and Evidence Checklist - Validation Management

This appendix page provides reusable evidence guidance for Validation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: VAL-EX-040. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.41: Detailed Example and Evidence Checklist - Management Review

This appendix page provides reusable evidence guidance for Management Review. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: MAN-EX-041. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.42: Detailed Example and Evidence Checklist - Regulatory Commitments

This appendix page provides reusable evidence guidance for Regulatory Commitments. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: REG-EX-042. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.43: Detailed Example and Evidence Checklist - Deviation Management

This appendix page provides reusable evidence guidance for Deviation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DEV-EX-043. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.44: Detailed Example and Evidence Checklist - CAPA Management

This appendix page provides reusable evidence guidance for CAPA Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CAP-EX-044. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.45: Detailed Example and Evidence Checklist - Change Control

This appendix page provides reusable evidence guidance for Change Control. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CHA-EX-045. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.46: Detailed Example and Evidence Checklist - Document Management

This appendix page provides reusable evidence guidance for Document Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DOC-EX-046. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.47: Detailed Example and Evidence Checklist - Training Management

This appendix page provides reusable evidence guidance for Training Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: TRA-EX-047. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.48: Detailed Example and Evidence Checklist - Risk Management

This appendix page provides reusable evidence guidance for Risk Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: RIS-EX-048. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.49: Detailed Example and Evidence Checklist - Audit Management

This appendix page provides reusable evidence guidance for Audit Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: AUD-EX-049. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.50: Detailed Example and Evidence Checklist - Complaint Management

This appendix page provides reusable evidence guidance for Complaint Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: COM-EX-050. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.51: Detailed Example and Evidence Checklist - Supplier Quality

This appendix page provides reusable evidence guidance for Supplier Quality. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: SUP-EX-051. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.52: Detailed Example and Evidence Checklist - Nonconformance

This appendix page provides reusable evidence guidance for Nonconformance. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: NON-EX-052. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.53: Detailed Example and Evidence Checklist - Equipment and Calibration

This appendix page provides reusable evidence guidance for Equipment and Calibration. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: EQU-EX-053. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.54: Detailed Example and Evidence Checklist - Validation Management

This appendix page provides reusable evidence guidance for Validation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: VAL-EX-054. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.55: Detailed Example and Evidence Checklist - Management Review

This appendix page provides reusable evidence guidance for Management Review. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: MAN-EX-055. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.56: Detailed Example and Evidence Checklist - Regulatory Commitments

This appendix page provides reusable evidence guidance for Regulatory Commitments. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: REG-EX-056. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.57: Detailed Example and Evidence Checklist - Deviation Management

This appendix page provides reusable evidence guidance for Deviation Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DEV-EX-057. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.58: Detailed Example and Evidence Checklist - CAPA Management

This appendix page provides reusable evidence guidance for CAPA Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CAP-EX-058. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.59: Detailed Example and Evidence Checklist - Change Control

This appendix page provides reusable evidence guidance for Change Control. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: CHA-EX-059. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

Appendix A.60: Detailed Example and Evidence Checklist - Document Management

This appendix page provides reusable evidence guidance for Document Management. Use it during formal UAT, OQ, or PQ execution.

Business Example

Example record: DOC-EX-060. A user creates a representative record, routes it through the configured Flowable workflow, performs required reviews, applies electronic signature where required, verifies audit trail, and confirms dashboard/report impact.

Evidence Checklist

Evidence Item	What to Capture	Pass Criteria
Record Header	Record number, status, creator, created date.	Header data is accurate and immutable after controlled state.
Workflow History	All tasks, assignees, dates, decisions.	Flow matches approved BPMN.
Role Permissions	Allowed and denied actions.	Only authorized users can act.
Attachments	Uploaded/downloaded evidence.	GCS link works and access is controlled.
Audit Trail	Create/update/status/signature events.	Complete who/what/when/old/new/reason.
E-signature	Meaning, signer, date/time, re-authentication.	Signature manifestation is visible.
Reports	Dashboard and exported report.	Metrics include the test record correctly.

Negative Test Example

Login with a user who does not have approval permission. Open the assigned record and attempt to approve through UI and API. Expected result: approval action is hidden or disabled in UI; API returns unauthorized/forbidden; no workflow transition occurs; security/audit event is logged if configured.

16. Final Acceptance Criteria

The Pharma QMS is ready for production when all critical end-to-end flows pass, all roles are verified, all records generate audit trails, e-signatures are enforced for approvals, reports are accurate, integrations are tested, and validation deliverables are approved.

- No unresolved critical or high defects.
- All workflow definitions deployed and version-controlled.
- All user roles tested with positive and negative cases.
- Audit trail and e-signature tests passed.
- End-to-end Deviation -> CAPA -> Change Control -> Document -> Training flow passed.
- Risk, Audit, Complaint, Supplier, Equipment, Validation, Management Review and Regulatory Commitment flows passed.
- IQ/OQ/PQ executed or mapped to customer validation approach.
- Production checklist approved by QA, IT, Validation, and Business Owner.